The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 December 1992

MIL-S-19500/394B
15 September 1992
SUPERSEDING
MIL-S-19500/394A
15 October 1972

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, MATCHED QUAD, FREE WHEEL, 1N6107 LAVIN AND JAMFET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, matched quad, discrete zone to switching circuits. Two levels of product assurance are provided for each device type as specified in MIL-S-19500. The "M" suffix device shall be from individually matched quads. Four discrete shall be shipped in a separate container (see 5.1).

1.2 Physical dimensions. See figures 1 and 2.

1.3 Maximum ratings. (T_A = +25°C unless otherwise specified).

Type	V_R	I_O	I_{FSM}			P_{RM}
			$t_1 \leq$		$t_2 \leq$	
	V dc	mA dc	mA	W	A	mW
1N6107	80	75	1125	500	8	250

1/ Derate linearly 3.0 mW/°C from +25°C to +150°C.
Storage temperature: T_{stg} = -65°C to +175°C.
Operating temperature: T_J = -65°C to +150°C.
Thermal impedance (junction-to-ambient): 0 °C/W.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be useful in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDS, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 5961

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3.4 Polarity of terminal and characteristics (each device).

Symbol	V_{CE}	V_{CE}	V_{CE}	V_{CE}	I_{CBO}
	I_C = 100 μA dc	I_C = 1 mA dc	I_C = 10 mA dc	I_C = 50 mA dc	V_{CB} = 75 V dc
	V dc	V dc	V dc	V dc	nA dc
Min	440	450	520	700	---
Max	650	770	870	1170	50

Symbol	I_{CEX}	I_{CEX}	C_{ob}	h_{FE}
	V_{CE} = 75 V dc	V_{CE} = 60 V dc V_{EB} = 1.5 V dc	V_{CB} = 5 V dc I_E = 0 mA dc	I_C = 10 mA dc V_{CE} = 10 V dc Pulsed (see 4.5.1)
	nA dc	nA dc	pF	---
Min	---	---	---	40
Max	5	25	60	200

1/ The reverse-recovery time (breakdown diode), condition A, of MIL-STD-750, method 4031, is a suitable equivalent and exceeds three times the t_{rr} limit.

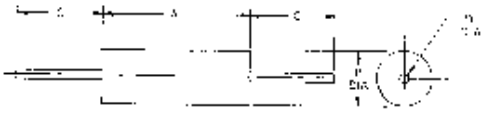
3.5 Electrical characteristics, matched pair. Each pair has the following characteristics in addition to those specified in 3.4.

Symbol	Δh_{FE}	ΔV_{BE}
	I_C = 0.1 mA to 1 mA dc V_{CE} = 10 V dc	I_C = 10 mA to 20 mA dc V_{CE} = 10 V dc
	---	mV dc
Min	---	---
Max	20	20

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2. APPLICABLE DOCUMENTS

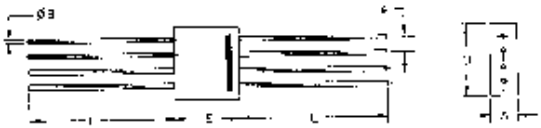
2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the DoDISS (Publication Index of Specifications and Standards (Publication Index of Specifications and Standards (Publication Index of Specifications and Standards

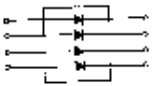


Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.720	.920	18.28	23.38	
B3	.065	.110	1.65	2.79	5
C	1.920	1.980	48.10	50.10	
g3	.271	.273	6.88	6.93	3, 4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. The specified test diameter applies to the zone between .250 (6.35 mm) and 1.000 (25.40 mm) from one shank end. Outside of this zone the test diameter is not controlled.
4. Both leads shall be within one specified orientation.
5. The minimum body diameter shall be maintained over its full length or area.

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CONNECTION DIAGRAM

Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.182	.188	4.62	4.78
ØB	.018	.021	0.46	.53
L	.487	.512	12.37	13.00
E	.425	.455	10.80	11.56
W	.100	.112	2.54	2.84
	.900	1.000	22.86	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. [illegible] applies to all leads.
4. [illegible] physical dimensions [illegible]

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. The qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 table III, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be accepted.

Screen (see table III of MIL-S-19500)	Measurement	
	JANTX and JANTXV levels	
11	V_{BR} and I_{BR}	at
12	MIL-STD-750, method 1750 case condition A, t = 10 cycles I_{BR} or ΔI_{BR}	
13	Subgroup 2 of table I herein. V_{BR} and I_{BR}: ΔV_{BR} = ±5% (max) ΔI_{BR} = ±100% or ±0.100 percent from initial value, whichever is greater.	

1/ The measurements shall not be indicative of an open condition.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = +125°C, V_R = 80 percent of rated V_R, 160 hours minimum.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Hard-point electrical measurements shall be in accordance with the applicable steps of table V herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVC (JANTX and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVC (JANTX and JANTXV) of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I herein.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIA of MIL-S-19500, and table IV herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I herein.

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TABLE I. Group A inspection.

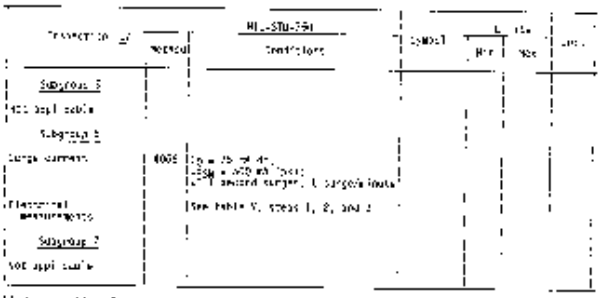
Inspection 1/	Method	Conditions	Symbol	Limits		Unit
				Min	Max	
Subgroup 2						
Visual and mechanical examination	2071					
Subgroup 3						
Forward voltage (each diode)	4011	I_F = 100 mA dc	V_{F1}	400	550	mV dc
Forward voltage (each diode)	4011	I_F = 1 mA dc	V_{F2}	.564	.674	V dc
Forward voltage (each diode)	4011	I_F = 10 mA dc	V_{F3}	.670	.810	V dc
Forward voltage (each diode)	4011	I_F = 30 mA dc	V_{F4}	.755	1.00	V dc
Reverse current (each diode)	4016	V_R = 50 V dc, cc method	I_{R1}		50	nA dc
Reverse current (each diode)	4016	V_R = V_{RM} dc, dc method	I_{R2}		0	µA dc
Reverse recovery time (each diode)	4031	Condition A; I_F = I_R = 10 mA dc; Recovery to I_R = 1 mA dc	t_{rr}		4	ns
Voltage match	4011	I_F = 0.1 µa to 10 mA µa T_A = +25°C to +125°C	ΔV_{F1}		10	mV dc
Voltage match	4011	I_F = 10 µa to 30 mA µa T_A = +25°C to +125°C	ΔV_{F2}		40	mV dc
Subgroup 4						
High temperature operation reverse current (each diode)	4016	T_A = +150°C; V_R = 50 V dc; cc method	I_{R3}		150	µA dc
Low temperature operation	4221	T_A = -55°C; I_F = 10 mA dc	V_{F3}	.522	.700	V dc
Subgroup 5						
Capacitance (each diode)	4001	V_R = 0 V dc, f = 1 MHz	C_T		4	pF
Reverse recovery time (each diode)	4031	Condition A or B; I_F = I_R = 10 mA dc; Recovery to I_R = 1 mA dc	t_{rr}		4	ns

See footnote at end of table.

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MIL-STD-1750A (USAF)

TABLE I. Cross reference - mnemonic, [illegible]



1/ For sampling data, see MIL-[illegible]

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TABLE II. Group A inspection for CA4, CB4TC, and CA4TXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	
Hermetic seal a. Fine leak b. Gross leak	1071	
Electrical measurements		See table I, steps 1, 2, and 3
Subgroup 3 2/		
Steady state operation life (seven diode)	1026	T_A = 25 ±3 dc, V_R = 80 V(pk) (f = 50 Hz
Electrical measurements		See table I, steps 1, 2, 3, 4, and 5
Subgroup 4		
Shock (internal visual (design verification)	2016	
Subgroup 5		
Not applicable		
Subgroup 6		
High-temperature life (nonoperating)	1031	T_A = 150°C
Electrical measurements		See table I, steps 1, 2, 3, 4, and 5

1/ For sampling plan, see MIL-S-19500.
2/ Devices from this group are not shippable.

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TABLE III. Group A inspection.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figures 1 and 2
Subgroup 2 2/		
Thermal shock (glass strain)	1056	
Lead integrity	2036a	Test condition A, weight = 4 pounds, t = 15 s
Moisture resistance	1021	Omit initial conditioning
Electrical measurements		See table V, steps 1, 2, and 3
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table V, steps 1, 2, and 3
Subgroup 4		
Solderability (terminals)	2026	
Subgroup 5		
Not applicable		

See footnotes at end of table.

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MIL-S-19500/394E

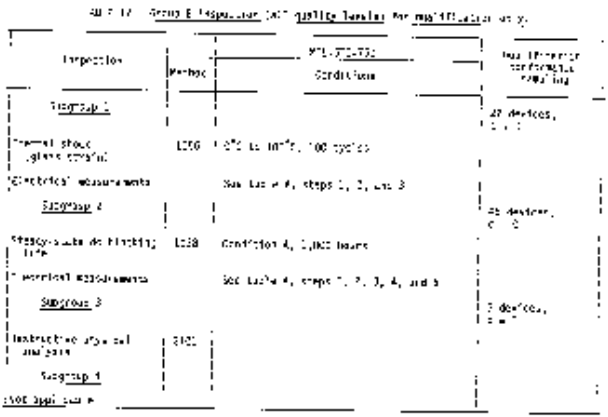
TABLE III. Group C inspection (non-JANTXV).

Inspection	MIL-STD-750	
	Method	Conditions
Subgroup 6 1/		
Steady-state operation life test (high temperature)	1026	T_J = 175 °C, V_R = 80 percent of $V_{(BR)DSS}$ minimum, V_{GS} = 20 V dc, 1,000 hours
Electrical measurements		See table V, steps 1, 2, 3, 4, and 5

1/ For sampling plan, see MIL-S-19500.
2/ Devices from this subgroup are not shippable

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METHOD 5004.8



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TABLE A. Group A, B, C, and E electrical end-point measurements.

Line	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 50 mA dc	V_{F2}	.75	1.0	V dc
2.	Reverse current	4016	DC method, V_R = 50 V dc	I_{R1}		5.	nA dc
3.	Capacitance	4001	V_R = 0, f = 1.0 MHz	C_T		2.0	pF
4.	Forward voltage	4011	I_F = 10 mA dc	ΔV_{F2}		±05	mV dc
5.	Reverse current	4016	DC method, V_R = 50 V dc	ΔI_{R1}		20% of initial value or ±10 nA dc, whichever is greater	

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4.5 Methods of inspection. Methods of inspection shall be as specified in applicable marking and as specified.

4.6 Storage recovery time. Storage recovery time shall be in accordance with MIL-STD-750, method 4077, condition B, if a fully assembled device is supplied.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-STD-1285.

5.2 Preservation for delivery of the device. The device shall be packaged in a manner that will maintain shelf-packed integrity during shipping and handling.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-STD-1690 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead times may be specified (see 3.1).

6.3 Substitution information. Devices covered by this specification are identified by the manufacturer's and user's Part or Identifying Number (PIN). This PIN is the way devices that manufacturer's PIN's are suitable as a substitute for one or more PIN.

Military PIN	Manufacturer's CAGE code	Manufacturer and home PIN

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6.4 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue due to the extensiveness of the changes.

OPERATIONS MATERIAL

Commitments:
Army - ER
Navy - EC
Air Force - LT
NASA - MR

Review activities
Army - AR, MD
Navy - OS, SH
Air Force - 19, 85, 99
DLA - ES

User activities
Army - AM
Navy - AS, FG, MI
Air Force - 23, ES

Preparing activity:
Navy - YD

Agent:
DLA - ES

(Project 5945-1105)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER M-38510/334A	2. DOCUMENT DATE (YYMMDD) 901215

3. DOCUMENT TITLE
Semiconductor Device, Diode, Silicon, Epitaxial Gold, 1N4307, 1N4307R, JAN, JANTX and JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	c. ADDRESS (Street, City, State, ZIP Code)	d. TELEPHONE (Include Area Code)
[illegible]	(1) Commercial (2) AUTOVON (If applicable)	[illegible]	(1) Commercial (2) AUTOVON
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